MARINE AUTOMATION PORTFOLIO · PROJECT 01

SMART EXHAUST GAS CLEANING SYSTEM (EGCS)

Automated Closed-Loop Scrubber Control
for Marine Emissions Compliance

TECHNICAL PROJECT MANUAL · REV. A

By Sipho Lucky Sibanda

PLC Platform: Siemens S7-1500 (SCL) / CODESYS-portable Structured Text
Simulation & Portfolio Engineering Build · Not for Shipboard Deployment

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FRONT MATTER

Document Control & Disclaimer

This document is a self-authored technical project manual produced as part of a personal engineering portfolio. It describes the design, control philosophy, and simulated validation of an Exhaust Gas Cleaning System (EGCS) scrubber controller, built to demonstrate PLC programming, closed-loop control design, and marine regulatory awareness for automation, controls, and marine engineering roles.

The system described here was developed and tested in simulation only (PLCSIM-style forcing of inputs and desktop review), using publicly available regulatory references for realism. No part of this project has been installed, commissioned, or verified on physical shipboard hardware, and it must not be treated as a certified or classed automation package.

SAFETY CRITICAL — Portfolio project, not a certified design
Figures, setpoints, and instrumentation ranges in this manual are engineering-realistic but illustrative. A real EGCS installation requires class society approval (e.g. DNV, ABS, Lloyd's Register), manufacturer-specific I/O, and a full HAZOP / FMEA study before any of this logic could be considered for actual use.

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FRONT MATTER

Executive Summary

Modern marine emissions regulation forces a choice on shipowners: burn expensive low-sulphur fuel everywhere, or install an exhaust gas cleaning system and keep burning cheaper heavy fuel oil while scrubbing SOx out before the exhaust reaches atmosphere. This project simulates the automation layer of that second option — specifically, the closed-loop control that keeps a wash-water scrubber both chemically compliant and legally compliant at the same time.

Two control problems sit on top of each other here. The first is a classic process control problem: hold discharge water pH above a regulatory minimum by modulating an alkaline dosing pump, using a PID loop reacting to a live pH sensor. The second is a regulatory/positional problem: no matter how chemically compliant the wash water is, it must never be discharged while the vessel is inside an Emission Control Area. That second condition can't be tuned away with a PID loop — it has to be a hard, unconditional interlock sourced from the vessel's navigation system.

The result is FB_EGCS_ScrubberControl, an IEC 61131-3 Structured Text function block that combines a tuned PID loop, hard chemical trips, a GPS/ECDIS-driven regulatory lockout, fail-safe sensor-fault handling, and rolling emissions logging — paired with a live Engine Control Room HMI mockup that visualises all of it, including the lockout event actually triggering.

This manual documents the regulatory context, architecture, hardware assumptions, full I/O list, control philosophy, complete annotated code, HMI design, alarm philosophy, and the functional test procedure used to validate the logic in simulation.

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EGCS SCRUBBER CONTROL

Regulatory & Marine Context

International shipping's air emissions are governed by MARPOL Annex VI, the International Maritime Organization's convention on the prevention of air pollution from ships. The rule that matters most for this project is the sulphur cap introduced under the IMO 2020 regulations: a global limit of 0.50% m/m sulphur content in marine fuel, tightened to 0.10% m/m inside designated Emission Control Areas (ECAs) such as the Baltic Sea, the North Sea, and the North American coastline.

A vessel has three ways to meet this: burn compliant low-sulphur fuel everywhere, burn an alternative fuel such as LNG, or keep burning cheaper heavy fuel oil and clean the exhaust before it reaches the funnel using an Exhaust Gas Cleaning System (EGCS), commonly called a scrubber. Open-loop and hybrid scrubbers spray seawater or an alkaline-dosed wash water through the exhaust stream; the wash water absorbs SOx, dropping its own pH in the process.

The wash-water problem

Cleaning the air creates a water compliance problem. The IMO's 2015 EGCS Guidelines (MEPC.259(68)) set limits on the wash water itself before it can be discharged overboard: a minimum discharge pH (commonly implemented at 6.5, measured at the overboard discharge point), a maximum pH depression relative to intake water, and monitoring of PAH (polycyclic aromatic hydrocarbon) proxy values and turbidity in more complete systems. This project focuses on the two variables most directly tied to the control loop: discharge pH and exhaust SOx concentration.

Emission Control Areas and discharge bans

Separately from water chemistry, many jurisdictions restrict or ban scrubber discharge outright within port limits, internal waters, or specific ECAs — regardless of how clean the wash water is. This is a purely positional rule: if the vessel is inside a restricted zone, discharge is not permitted. That distinction is the reason this project's control logic treats chemistry compliance and positional compliance as two independent interlocks rather than one blended "compliant/non-compliant" flag.

ENGINEERING NOTE — Why two interlocks, not one

It would be simple to fold pH, SOx, and ECA status into a single boolean and call it "compliant". Real systems don't do that, because a positional interlock has zero tolerance and no PID-style approach curve — the moment the vessel crosses the ECA boundary, discharge must stop immediately. Chemistry compliance, by contrast, is something the control loop actively works towards. Keeping them separate in the logic means a slow chemistry excursion can never be masked by "well, we're outside the ECA", and a hard position trip can never be delayed by "well, the pH is fine".

Record-keeping

MARPOL Annex VI compliance is also a paperwork exercise: class societies and port state control expect continuous emissions and discharge records. The control logic in this project reflects that by maintaining a rolling log of pH, SOx, dosing rate, discharge state, and ECA status — the PLC-side equivalent of what would, on a real vessel, stream into a SCADA historian for audit.

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EGCS SCRUBBER CONTROL

System Architecture & Process Overview

The process flow is linear: exhaust gas from the main and/or auxiliary engines enters the scrubber tower, where a wash-water spray deck strips out SOx before the cleaned gas exits to the funnel. The wash water itself is then routed past a pH/SOx analyser pair, which feeds the PLC; the PLC in turn drives an alkaline dosing pump and gates the overboard discharge valve. A GPS/ECDIS position feed runs in parallel as an independent regulatory input.

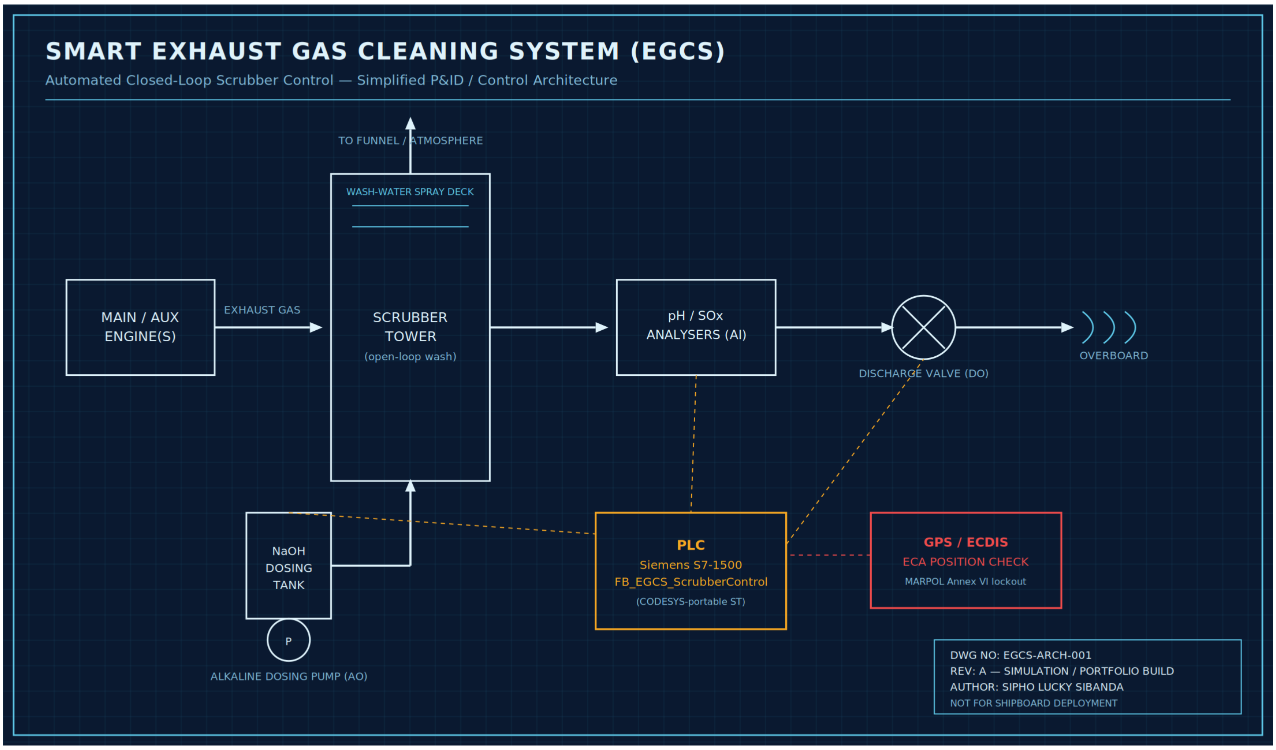

Figure 2.1 — Simplified process and control architecture. Solid lines represent process piping; dashed lines represent PLC signal paths.

Control hierarchy

- Field layer — pH probe, SOx analyser, exhaust temperature RTD, dosing pump VFD, discharge valve actuator, GPS/ECDIS position feed.
- Control layer — a single PLC function block, FB_EGCS_ScrubberControl, hosted on a Siemens S7-1500 (or CODESYS-based marine controller), running the PID loop, interlocks, and logging.
- Supervisory / HMI layer — an Engine Control Room display giving the watch-keeping engineer live process values, alarm status, and an explicit ECA lockout indication.

Design intent

The function block is deliberately self-contained: all setpoints, tuning constants, and logging live inside it, and it exposes a small, clean input/output interface. On a real vessel this would sit inside a larger Integrated Automation System (IAS) alongside engine, power management, and alarm-monitoring functions — the interface is kept narrow specifically so it could be dropped into that kind of system without dragging unrelated logic with it.

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EGCS SCRUBBER CONTROL

Hardware & Instrumentation Specification

The control logic in this project is platform-portable, but it was written against a concrete hardware assumption so that setpoints, scaling, and timing are realistic rather than arbitrary. The reference platform is a Siemens S7-1500 CPU programmed in SCL through TIA Portal, chosen because it's one of the most common PLC families in marine and industrial automation and its Structured Text is close enough to the CODESYS/IEC 61131-3 standard to port with minor changes.

Component	Representative Spec	Role
PLC CPU	Siemens SIMATIC S7-1500 (e.g. CPU 1515-2 PN)	Executes FB_EGCS_ScrubberControl
pH Analyser	4–20mA, 0–14 pH, glass electrode or ISFET	Discharge water pH feedback
SOx Analyser	4–20mA, 0–3000ppm, NDIR or electrochemical	Exhaust SOx concentration
Exhaust Temp.	Pt100 RTD, 0–450°C	Process monitoring / thermal protection
Dosing Pump	VFD-driven metering pump, 4–20mA speed ref.	Alkaline (NaOH) dosing
Discharge Valve	Motorised or pneumatic, 24VDC digital control	Overboard discharge permissive
Position Feed	NMEA 0183/2000 GPS via ECDIS/nav. interface	ECA boundary determination
I/O Coupling	Distributed I/O or direct rack-mounted modules	Field signal termination

Why these instrument choices

4–20mA analogue signalling is used for the two safety/compliance-critical measurements (pH, SOx) because it is inherently loop-fault detectable — a broken wire or dead transmitter reads as 0mA, well outside the live 4–20mA band, which is exactly the condition Sensor_Fault is designed to catch in the control logic (Chapter 6). A purely digital or fieldbus-only sensor without that kind of wire-break detection would need an equivalent watchdog implemented elsewhere.

NOTE — Signal scaling in TIA Portal

In a real S7-1500 project, the raw 0–27648 analogue input word from a 4–20mA channel is scaled to engineering units (0.00–14.00 pH, 0–3000 ppm) using the NORM_X / SCALE_X function blocks before it ever reaches FB_EGCS_ScrubberControl — the function block itself expects clean, already-scaled REAL values as shown in the I/O list.

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EGCS SCRUBBER CONTROL

I/O List

The table below is the working I/O list for FB_EGCS_ScrubberControl, in the same format used on the project's FAT/commissioning documentation (see also docs/IO_List.md in the repository, and Appendix A of this manual).

Inputs

Tag	Description	Signal	Range / Units
AI_pH_Feedback	Discharge water pH probe	4–20mA	0.00–14.00 pH
AI_SOx_ppm	Exhaust SOx analyser	4–20mA	0–3000 ppm
AI_ExhaustTemp	Exhaust gas temperature	Pt100 RTD	0–450°C
DI_GPS_InECA	Vessel-in-ECA flag (nav. system)	24VDC digital	0/1
DI_System_Enable	Master enable (bridge/ECR HMI)	24VDC digital	0/1
DI_Sensor_Fault	Combined pH/SOx comms fault	24VDC digital	0/1

Outputs

Tag	Description	Signal	Range / Units
AO_DosingPump_Speed	Alkaline dosing pump VFD ref.	4–20mA	0–100%
DO_WashWaterDischargeValve	Discharge valve permissive	24VDC digital	0=Closed / 1=Open
DO_Alarm_HighSOx	High SOx alarm	24VDC digital	—
DO_Alarm_LowpH	Low pH alarm	24VDC digital	—
DO_Alarm_ECA_Lockout	ECA lockout indicator	24VDC digital	—
DO_Alarm_SensorFault	Sensor/comms fault indicator	24VDC digital	—

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EGCS SCRUBBER CONTROL

Control Philosophy: PID Loop & Interlocks

Three distinct control mechanisms operate on top of each other in this system, each suited to a different kind of requirement: a continuous PID loop for the process variable that genuinely needs tuning, hard threshold trips for conditions that must never be allowed to drift, and an unconditional interlock for the one input that has nothing to do with process chemistry at all.

5.1 The pH control loop

Discharge pH is controlled by a standard single-loop PID, adjusting the alkaline dosing pump's speed reference (0–100%) to hold pH at the setpoint of 6.50. Proportional gain and integral time were tuned conservatively (K_p = 8.0, T_i = 20s, no derivative action) because wash-water pH in a real scrubber has meaningful transport delay between the dosing point and the analyser — aggressive tuning on a delayed loop like this tends to ring rather than settle.

Parameter	Value	Rationale
Setpoint (pH)	6.50	IMO MEPC.259(68) minimum discharge pH
Kp	8.0	Moderate gain — avoids overshoot on a lagging loop
Ti	20 s	Integral action tuned to typical wash-water transport delay
Td	0 s	No derivative term — sensor noise would be amplified
Output limits	0–100%	Full VFD speed range on the dosing pump

5.2 Hard trips vs. soft alarms

Below the PID's normal working range sits a hard trip: if pH falls to or below 6.00 (pH_LowLow), the loop is bypassed entirely and the dosing pump is forced to 100% regardless of what the PID output would otherwise say. This exists because a PID loop, by design, approaches its setpoint gradually — that's the wrong behaviour once pH has already dropped into territory that risks being out of compliance right now. A soft alarm (Alarm_LowpH) is raised any time pH is below setpoint at all, well before the hard trip point, so the watch engineer sees the trend developing.

5.3 The ECA interlock is not a PID input

The GPS/ECDIS ECA flag never enters the PID loop and never modulates anything gradually. It is evaluated as a single Boolean condition that, when true, unconditionally removes the discharge valve's permissive — independent of how compliant the water chemistry is. This mirrors how these systems are specified in practice: position-based discharge bans are treated as zero-tolerance conditions, not soft constraints to be optimised against.

> **REGULATORY NOTE — Why not just close the valve slowly on ECA entry?**
> It's tempting to add a ramp-down so the valve doesn't slam shut. That would be the wrong engineering choice here: any discharge after the vessel is confirmed inside a restricted zone is a regulatory breach regardless of duration. The interlock is intentionally abrupt because the underlying rule is abrupt.

5.4 Fail-safe default

Any sensor or communications fault (Sensor_Fault) forces the system to its safest state — dosing pump off, discharge valve closed — on the same scan it's detected, ahead of every other evaluation in the function block. The system does not attempt to keep discharging on stale or assumed-good sensor values.

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EGCS SCRUBBER CONTROL

PLC Logic Walkthrough

This chapter walks through FB_EGCS_ScrubberControl section by section. The full, unbroken listing is reproduced in Appendix B; what follows here is the same code split into its seven logical stages with the reasoning behind each one.

6.1 Sensor fault handling comes first

The function block checks for sensor/comms faults before anything else runs. This ordering is deliberate — every other rung in the block assumes the input values it's reading are trustworthy, so the fail-safe check has to short-circuit everything downstream of it, not sit alongside it.

```
IF Sensor_Fault THEN
    Alarm_SensorFault := TRUE;
    AlkalineDosingPump_Speed := 0.0;
    WashWaterDischargeValve := FALSE;
    SystemStatus := 'SENSOR FAULT';
    RETURN;
ELSE
    Alarm_SensorFault := FALSE;
END_IF
```

Listing 6.1 — Fail-safe fault handling, evaluated first on every scan.

6.2 The ECA interlock is evaluated independently

DischargePermitted is calculated directly from the enable/ECA/fault state, kept separate from the chemistry checks that come later. This is what lets Chapter 5's "interlock, not PID input" philosophy actually hold up in code — there's no path through this logic where a favourable pH reading can compensate for being inside an ECA.

```
Alarm_ECA_Lockout := GPS_InECA;
DischargePermitted := System_Enable AND NOT GPS_InECA AND NOT Sensor_Fault;
```

Listing 6.2 — Regulatory position interlock.

6.3 PID loop with a hard low-pH trip

The PID instance drives dosing pump speed under normal conditions; the hard trip overrides it only when pH has already dropped to the LowLow threshold, forcing full dosing rather than waiting for the loop to catch up.

```
IF System_Enable THEN
    PID_Inst(SP := pH_Setpoint, PV := pH_Feedback, KP := Kp, TI := Ti_s, TD := Td_s,
             MANUAL := FALSE, LMN_HLM := 100.0, LMN_LLM := 0.0);
    PID_Output := PID_Inst.LMN;

    IF pH_Feedback <= pH_LowLow THEN
        AlkalineDosingPump_Speed := 100.0;
    ELSE
        AlkalineDosingPump_Speed := PID_Output;
    END_IF
ELSE
    AlkalineDosingPump_Speed := 0.0;
    PID_Inst(MANUAL := TRUE, LMN := 0.0);
END_IF
```

Listing 6.3 — PID dosing control with hard trip override.

6.4 Alarms, valve permissive, and status text

Soft alarms are simple threshold comparisons. The discharge valve permissive is the single point where every condition — enable, position, pH, SOx, sensor health — has to agree before the valve is allowed open; SystemStatus exists purely to drive a human-readable HMI banner from the same underlying booleans.

```
Alarm_LowpH := (pH_Feedback < pH_Setpoint) AND System_Enable;
Alarm_HighSOx := (SOx_ppm > SOx_Limit_ppm);

WashWaterDischargeValve := DischargePermitted
                           AND (pH_Feedback >= pH_Setpoint)
                           AND (SOx_ppm <= SOx_HighHigh);

IF NOT System_Enable THEN SystemStatus := 'STANDBY';
ELSIF GPS_InECA THEN SystemStatus := 'ECA - DISCHARGE LOCKED';
ELSIF Alarm_HighSOx OR Alarm_LowpH THEN SystemStatus := 'OUT OF COMPLIANCE';
ELSE SystemStatus := 'COMPLIANT - RUNNING';
END_IF
```

Listing 6.4 — Alarm evaluation and the discharge valve permissive.

6.5 Rolling emissions log

A 60-second retentive timer writes a snapshot of pH, SOx, dosing rate, discharge state, and ECA status into a circular buffer. On real hardware this record would stream to a SCADA historian rather than live in PLC memory indefinitely; the circular buffer here stands in for that at the function-block level.

```
LogTimer(IN := System_Enable, PT := LogTimer_PT);
IF LogTimer.Q THEN
    LogTimer(IN := FALSE);
    EmissionLog[LogIndex].pH_Value := pH_Feedback;
    EmissionLog[LogIndex].SOx_ppm := SOx_ppm;
    EmissionLog[LogIndex].DoseRate_Pct := AlkalineDosingPump_Speed;
    EmissionLog[LogIndex].DischargeActive := WashWaterDischargeValve;
    EmissionLog[LogIndex].InECA := GPS_InECA;
    LogIndex := LogIndex + 1;
    IF LogIndex > 999 THEN LogIndex := 0; END_IF
END_IF
```

Listing 6.5 — Rolling MARPOL Annex VI emissions record.

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EGCS SCRUBBER CONTROL

HMI Design & Operator Experience

The HMI mockup (hmi/index.html in the repository) simulates the screen a watch engineer would see in the Engine Control Room: live process values, dosing pump and valve state, a simplified process schematic, an alarm/event log, and position/compliance data. It runs entirely client-side with no dependencies, and periodically drives the vessel in and out of an ECA so the lockout state can actually be observed rather than just described.

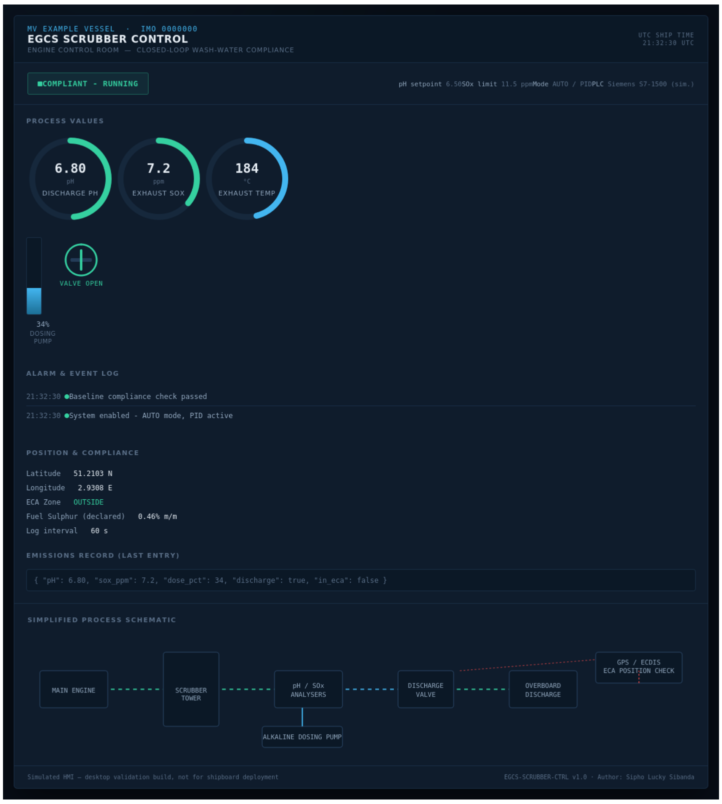

Figure 7.1 — Engine Control Room HMI, showing the compliant/running state.

Design decisions

- Dark theme — deliberately chosen, not a default: engine control rooms and bridge repeaters are frequently viewed during night watches, and a dark, low-glare interface is standard practice on real marine HMIs (e.g. Kongsberg K-Chief, Siemens SIMATIC WinCC marine panels).
- Status pill as the single source of truth — rather than making the operator cross-reference three separate readouts, one colour-coded pill states the system's overall condition in words, driven directly off the same SystemStatus logic as the PLC.
- The ECA banner is the signature element — it is intentionally the loudest thing on the screen when active, because it represents the one condition in this system with zero tolerance for delay or misreading.

- SVG arc gauges over CSS gradients — radial gauges are drawn as SVG stroke arcs rather than CSS conic-gradients, specifically so the interface renders identically across browser engines, including older ones — the same reasoning a real HMI runtime vendor would apply.

ENGINEERING NOTE — Engineering lesson worth keeping

An early build of this HMI used modern CSS (conic-gradient) and ES6 JavaScript (template literals, arrow functions). It looked correct in a modern browser but silently failed — blank gauges, empty alarm log — when rendered through an older WebKit-based engine used for automated screenshotting. The fix was to rewrite the interactive layer in plain ES5 and swap the gauges to SVG stroke arcs. The takeaway: an HMI that only works in one rendering engine is a real defect, not a cosmetic one — shipboard displays and vendor-specific panel PCs often run exactly the kind of older embedded browser this exposed.

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EGCS SCRUBBER CONTROL

Alarm Philosophy & Fail-Safe Design

Four distinct alarm/status conditions are raised by the function block, each mapped to a specific operator action rather than a generic "something's wrong" light:

Condition	Meaning	Expected Operator Response
Alarm_LowpH	Discharge pH below setpoint	Monitor trend; verify dosing pump is responding
Alarm_HighSOx	Exhaust SOx above compliance limit	Check scrubber performance / engine load
Alarm_ECA_Lockout	Vessel inside an Emission Control Area	Informational — no action; system self-manages
Alarm_SensorFault	pH/SOx sensor or comms failure	Dispatch engineer to inspect field instrument

Fail-safe defaults, summarised

- Loss of sensor signal → dosing pump off, discharge valve closed.
- Loss of system enable → dosing pump ramps to zero, discharge valve closes, status returns to STANDBY.
- ECA entry → discharge valve permissive removed immediately, no ramp or delay.
- No alarm in this system latches silently — every condition clears automatically once its underlying cause clears, which keeps the operator's picture of the plant honest in real time rather than requiring manual acknowledgement to "unstick" a resolved condition.

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EGCS SCRUBBER CONTROL

Testing, Commissioning & FAT Procedures

The function block was validated against ten functional test cases covering normal operation, boundary conditions, and fault handling — structured the way a Factory Acceptance Test (FAT) would be run before a real system moves to Site Acceptance Testing (SAT) on physical I/O. The full procedure, including a sign-off table, is in docs/Testing_Procedures.md; the test matrix is reproduced below.

#	Test Case	Expected Result
1	System start-up	Status moves STANDBY → COMPLIANT-RUNNING; pump ramps from 0%
2	pH closed-loop response	Pump speed increases smoothly, no oscillation >±5%
3	pH low-low hard trip	Pump snaps to 100%; Alarm_LowpH raised
4	High SOx alarm	Alarm_HighSOx raised; valve permissive drops above HighHigh
5	ECA GPS interlock	Valve closes within one scan; lockout status shown
6	ECA exit recovery	Normal logic resumes automatically, no manual reset
7	Sensor fault fail-safe	Pump 0%, valve closed, status = SENSOR FAULT
8	Fault recovery	Resumes without a bumped output on recovery
9	Emission logging cadence	New record every 60s; index wraps at 999→0
10	Manual disable	Pump to 0%, valve closes, status = STANDBY

NOTE — Testing without physical hardware

All ten cases were exercised by forcing input tags directly — the PLCSIM pattern for Siemens platforms, or an equivalent soft-PLC harness for CODESYS — and observing outputs. This is exactly how a controls engineer validates logic before it ever sees a cabinet, and it's a legitimate, recruiter-relevant way to demonstrate testing discipline without owning a scrubber skid.

9.1 Commissioning sequence (SAT-style, for reference)

The FAT above proves the logic. A real Site Acceptance Test on installed hardware would follow a broader sequence, included here to show the full path from simulated logic to a commissioned system — and because being able to describe that path is itself something recruiters in this field screen for.

Step	Activity	Exit Criteria
1	Cold loop checks	Every I/O point traced from field terminal to PLC tag; no crossed wiring
2	Instrument calibration	pH probe 2-point buffer calibration; SOx analyser zero/span verified
3	Actuator stroke test	Dosing pump VFD responds correctly to 0/50/100% reference; valve full travel timed
4	Interlock proving	ECA and sensor-fault interlocks physically forced and confirmed to trip within spec
5	Closed-loop tuning check	PID response observed on real process; Kp/Ti fine-tuned against Chapter 5 baseline
6	Alarm walk-down	Every alarm in Chapter 8 raised and cleared once, confirmed on HMI and class-required log
7	Endurance run	24-hour continuous run under normal load with no unexplained trips
8	Sign-off	Class surveyor and shipowner's engineer countersign the FAT/SAT record

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EGCS SCRUBBER CONTROL

Limitations, Real-World Deltas & Future Work

This project is explicit about the gap between a strong portfolio simulation and a certifiable shipboard system. Naming that gap accurately is itself part of the engineering — it's the difference between a project that looks impressive and one that demonstrates you understand what "production-ready" actually requires.

What a real installation would add

- Class society approval — DNV, ABS, or Lloyd's Register type-approval of the EGCS package and a documented HAZOP/FMEA on the control logic itself.
- Redundant sensing — a single pH/SOx analyser pair is a simplification; real systems often vote multiple sensors to avoid a single point of failure driving a false compliant reading.
- ECA boundary logic — this project treats GPS_InECA as a single input bit; a production system would run a live polygon check against IMO-published ECA charts, likely on a separate navigation-integration controller rather than inside the scrubber PLC.
- PAH and turbidity monitoring — full EGCS Guidelines compliance monitors additional wash-water parameters beyond pH and SOx, omitted here to keep the control loop legible for a portfolio piece.
- Cybersecurity hardening — network segmentation, signed firmware, and access control for the PLC and HMI, per IEC 62443 — a real theme across this whole portfolio series, developed in depth in the maritime cybersecurity project.

10.1 Hazard & safeguard register (HAZOP-style)

A full HAZOP is out of scope for a portfolio simulation, but naming the main hazards this control philosophy is actually defending against — and where the responsibility sits outside the PLC — is exactly the kind of thinking a HAZOP session would produce. A short version is reproduced below.

Hazard	Cause	Safeguard in This Design
Non-compliant discharge	pH sensor drifts high while actually low	Redundant sensing recommended (Ch.10); hard LowLow trip independent of PID
Discharge inside an ECA	GPS/ECDIS feed lost or stale	DI_Sensor_Fault path also covers loss of the position feed — fails to valve-closed
Over-dosing (alkaline waste)	PID integral windup on a stuck sensor	LMN_HLM/LMN_LLM clamps on the PID instance; fault path forces pump to 0% independently
Silent instrument failure	4-20mA loop reads a plausible mid-range value while faulted	Requires field-side loop diagnostics upstream of the PLC (see Ch.3) — noted as a residual risk

Hazard	Cause	Safeguard in This Design
Operator overrides interlock	Manual bypass switch fitted at cabinet	Not implemented in this design by choice — ECA lockout has no software or field bypass path

REGULATORY NOTE — Residual risk, stated plainly

The "silent instrument failure" row above is the honest limitation of this design: a 4-20mA signal that fails to a plausible mid-range value rather than to 0mA will not trip Sensor_Fault. Real installations close this gap with loop diagnostics at the analyser/transmitter level, or a second, independently-sourced measurement voted against the first — deliberately left as a named gap here rather than solved with an assumption that would make the design look safer than it is.

Where this project could go next

- Add a second-stage SOx feed-forward term, so a sudden engine load increase pre-empts the pH loop rather than waiting for the pH excursion to happen.
- Extend the emissions log to export as CSV for direct import into a reporting tool.
- Add a simple polygon-based ECA check driven by real ECA boundary coordinates, replacing the single simulated flag.

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APPENDIX

Appendix A — I/O Quick Reference

Condensed combined I/O reference for bench/desk use alongside the PLCSIM watch table.

Tag	Dir.	Type	Notes
AI_pH_Feedback	IN	REAL (4-20mA)	0.00-14.00 pH
AI_SOx_ppm	IN	REAL (4-20mA)	0-3000 ppm
AI_ExhaustTemp	IN	REAL (RTD)	0-450 C
DI_GPS_InECA	IN	BOOL	1 = inside ECA
DI_System_Enable	IN	BOOL	Master enable
DI_Sensor_Fault	IN	BOOL	1 = comms/sensor fault
AO_DosingPump_Speed	OUT	REAL (4-20mA)	0-100%
DO_WashWaterDischargeValve	OUT	BOOL	1 = open
DO_Alarm_HighSOx	OUT	BOOL	Soft alarm
DO_Alarm_LowpH	OUT	BOOL	Soft alarm
DO_Alarm_ECA_Lockout	OUT	BOOL	Informational
DO_Alarm_SensorFault	OUT	BOOL	Fail-safe trigger

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APPENDIX

Appendix B — Full Structured Text Listing

Complete, unedited listing of src/EGCS_ScrubberControl.st.

```
(*
=====================================================================================
  PROJECT : Smart Exhaust Gas Cleaning System (EGCS)
              Automated Closed-Loop Scrubber Control
  MODULE : FB_EGCS_ScrubberControl
  PLATFORM : IEC 61131-3 Structured Text (Siemens SCL / CODESYS-portable)
  AUTHOR : Sipho Lucky Sibanda
  CONTEXT : Alkaline-dosed wash water neutralises SOx absorbed from engine
              exhaust before discharge overboard or holding for treatment.
  REGULATORY BASIS (simulated against, for realism only):
    - MARPOL Annex VI, Regulation 14 : 0.50% m/m global / 0.10% m/m in ECAs
    - IMO MEPC.259(68) 2015 Guidelines: discharge pH >= 6.5
=====================================================================================
*)

TYPE ST_EmissionRecord :
STRUCT
    Timestamp : DT;
    pH_Value : REAL;
    SOx_ppm : REAL;
    DoseRate_Pct : REAL;
    DischargeActive : BOOL;
    InECA : BOOL;
END_STRUCT
END_TYPE
```

```
FUNCTION_BLOCK FB_EGCS_ScrubberControl
VAR_INPUT
    pH_Feedback : REAL;
    SOx_ppm : REAL;
    ExhaustTemp_C : REAL;
    GPS_InECA : BOOL;
    System_Enable : BOOL;
    Sensor_Fault : BOOL;
END_VAR

VAR_OUTPUT
    AlkalineDosingPump_Speed : REAL := 0.0;
    WashWaterDischargeValve : BOOL := FALSE;
    Alarm_HighSOx : BOOL := FALSE;
    Alarm_LowpH : BOOL := FALSE;
    Alarm_ECA_Lockout : BOOL := FALSE;
    Alarm_SensorFault : BOOL := FALSE;
    SystemStatus : STRING[20] := 'STANDBY';
END_VAR

VAR
    pH_Setpoint : REAL := 6.50;
    pH_LowLow : REAL := 6.00;
    SOx_Limit_ppm : REAL := 11.50;
    SOx_HighHigh : REAL := 14.00;

    PID_Inst : FB_PID;
    Kp : REAL := 8.0;
    Ti_s : REAL := 20.0;
    Td_s : REAL := 0.0;
    PID_Output : REAL;

    EmissionLog : ARRAY[0..999] OF ST_EmissionRecord;
    LogIndex : INT := 0;
    LogTimer : TON;
    LogTimer_PT : TIME := T#60S;

    DischargePermitted : BOOL;
END_VAR
```

```
// 1. SENSOR / COMMS FAULT HANDLING - fails to safe state
IF Sensor_Fault THEN
    Alarm_SensorFault := TRUE;
    AlkalineDosingPump_Speed := 0.0;
    WashWaterDischargeValve := FALSE;
    SystemStatus := 'SENSOR FAULT';
    RETURN;
ELSE
    Alarm_SensorFault := FALSE;
END_IF

// 2. REGULATORY / ECA GPS INTERLOCK
Alarm_ECA_Lockout := GPS_InECA;
DischargePermitted := System_Enable AND NOT GPS_InECA AND NOT Sensor_Fault;

// 3. CLOSED-LOOP pH CONTROL (PID)
IF System_Enable THEN
    PID_Inst(SP := pH_Setpoint, PV := pH_Feedback, KP := Kp, TI := Ti_s, TD := Td_s,
             MANUAL := FALSE, LMN_HLM := 100.0, LMN_LLM := 0.0);
    PID_Output := PID_Inst.LMN;

    IF pH_Feedback <= pH_LowLow THEN
        AlkalineDosingPump_Speed := 100.0;
    ELSE
        AlkalineDosingPump_Speed := PID_Output;
    END_IF
ELSE
    AlkalineDosingPump_Speed := 0.0;
    PID_Inst(MANUAL := TRUE, LMN := 0.0);
END_IF
```

```
// 4. ALARM EVALUATION
Alarm_LowpH := (pH_Feedback < pH_Setpoint) AND System_Enable;
Alarm_HighSOx := (SOx_ppm > SOx_Limit_ppm);

// 5. DISCHARGE VALVE PERMISSIVE
WashWaterDischargeValve := DischargePermitted
                           AND (pH_Feedback >= pH_Setpoint)
                           AND (SOx_ppm <= SOx_HighHigh);

// 6. SYSTEM STATUS TEXT (drives HMI banner)
IF NOT System_Enable THEN
    SystemStatus := 'STANDBY';
ELSIF GPS_InECA THEN
    SystemStatus := 'ECA - DISCHARGE LOCKED';
ELSIF Alarm_HighSOx OR Alarm_LowpH THEN
    SystemStatus := 'OUT OF COMPLIANCE';
ELSE
    SystemStatus := 'COMPLIANT - RUNNING';
END_IF
```

```
// 7. REGULATORY EMISSIONS LOGGING
LogTimer(IN := System_Enable, PT := LogTimer_PT);
IF LogTimer.Q THEN
    LogTimer(IN := FALSE);
    EmissionLog[LogIndex].Timestamp := DT#2026-01-01-00:00:00;
    EmissionLog[LogIndex].pH_Value := pH_Feedback;
    EmissionLog[LogIndex].SOx_ppm := SOx_ppm;
    EmissionLog[LogIndex].DoseRate_Pct := AlkalineDosingPump_Speed;
    EmissionLog[LogIndex].DischargeActive := WashWaterDischargeValve;
    EmissionLog[LogIndex].InECA := GPS_InECA;

    LogIndex := LogIndex + 1;
    IF LogIndex > 999 THEN
        LogIndex := 0;
    END_IF
END_IF

END_FUNCTION_BLOCK
```

Listing B.1 — Complete FB_EGCS_ScrubberControl source.

0C

APPENDIX

Appendix C — Glossary

Term	Meaning
EGCS	Exhaust Gas Cleaning System — a marine scrubber
ECA	Emission Control Area — a zone with tighter sulphur/NOx limits under MARPOL Annex VI
MARPOL Annex VI	The IMO convention chapter governing air pollution from ships
MEPC.259(68)	IMO guidelines specifying EGCS wash-water discharge criteria
PID	Proportional-Integral-Derivative — a standard closed-loop control algorithm
FB	Function Block — a reusable, encapsulated unit of PLC logic (IEC 61131-3)
SCL	Structured Control Language — Siemens' IEC 61131-3 Structured Text dialect
ECDIS	Electronic Chart Display and Information System — the vessel's navigation display
FAT / SAT	Factory / Site Acceptance Test — formal pre- and post-installation validation stages
HAZOP	Hazard and Operability study — a structured method for identifying process risks
I/O	Input/Output — the physical or logical signals a PLC reads and writes
VFD	Variable Frequency Drive — motor speed control used here on the dosing pump

CLOSING

About the Author

Sipho Lucky Sibanda is an automation and controls engineer building a multi-disciplinary portfolio spanning marine systems, industrial automation, biotech, and applied AI/PLC integration. This manual is the first in a series documenting each project in the Marine Automation Portfolio to the same standard: full PLC logic, I/O documentation, a live HMI, functional test procedures, and an honest account of what separates a strong simulation from a certifiable production system.

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End of document.